

# Anytime-Valid Conjoint Analysis: A Candidate-Gender Case Study

Supplementary case study for *Anytime-Valid Inference in Conjoint Experiments* (Molitor, Gosciak, and Lindon, 2026)

July 14, 2026

## 1 Introduction

The parent paper establishes anytime-valid e-values and confidence sequences (CSs) for the linear-probability (“AMCE”) model of conjoint analysis, and demonstrates their properties through simulation. This note complements those simulations, and the companion digital-camera case study, with a case study on a real political-science conjoint dataset: a discrete-choice survey in which voters were shown pairs of hypothetical political candidates and chose which one they would support. The substantive question motivating the underlying study – and the one this case study foregrounds – is whether a candidate’s gender affects voter support once other biographical attributes are held fixed. As in the companion case study, the goal is to show what the anytime-valid method looks like end-to-end on real data: how the confidence sequence and e-value for each candidate attribute, especially candidate gender, evolve as respondents accumulate, and how the final anytime-valid interval compares to the conventional fixed- $n$  interval computed from the same fitted model.

All code, data, and figures for this case study live in `candidate_gender_case_study/` and can be regenerated with `run_all.sh`; see the accompanying `README.md` for exact reproduction steps.

## 2 Data

We use the candidate-choice conjoint experiment of Teele, Kalla, and Rosenbluth (2018), fielded to study whether social-role expectations and traditionally-gendered biographical traits shape support for female political candidates. Respondents were shown pairs of hypothetical candidate biographies – differing in years of political experience, spousal status, occupation, number of children, age, and gender – and asked which candidate they would support. The outcome `winner` records whether a given candidate profile was the one chosen in its pairwise comparison; this is the AMCE outcome used throughout this case study, playing the same role as `chosen` in the companion digital-camera case study. Full column definitions and provenance are in `data/raw/data_dictionary.csv` and `data/raw/dataverse_README.txt`.

The full replication data span both a general-population voter sample and a sample of political elites/legislators, as well as a 2017 legislator-only replication wave fielded with a modified attribute set. To isolate a single clean experimental condition, comparable in spirit to the general-population conjoint samples used elsewhere in this paper, we restrict to the original 2014 wave of the general-population voter sample under the unmodified attribute set (`sample == "usa voter" & replication == 0`). The elite/ legislator sample and the 2017 replication are real parts of the original study but out of scope here; see the Discussion for these as natural extensions. After this

filter, the data comprise 2,105 respondents (clusters), each completing up to 3 candidate-comparison tasks, for 12,450 respondent-task-candidate rows and 6,225 valid pairwise comparisons in total.

Each candidate profile is described by five attributes, one-hot encoded in the raw data with no reference level omitted: years of political experience (0/1/3/8), spousal status (unmarried / married to a farmer / married to a doctor), occupation (teacher / lawyer / mayor / legislator), number of children (0/1/3), and age (29/45/65) – plus a sixth, binary attribute, the candidate’s gender (`orig_cand_female`), which is the attribute of central substantive interest for this case study. Because each group’s dummies sum to 1 in every row, we drop exactly one reference dummy per group before fitting (0 years, unmarried, teacher, 0 children, age 29) to avoid perfect collinearity with the intercept, leaving 13 non-intercept coefficients.

We treat each of the 2,105 respondents as a cluster, exactly as in the simulations in the main paper and in the companion camera case study, and fit

$$\begin{aligned} \text{winner} \sim & \text{orig\_1ys} + \text{orig\_3ys} + \text{orig\_8ys} + \text{orig\_FM\_sp} + \\ & \text{orig\_MD\_sp} + \text{orig\_law} + \text{orig\_may} + \text{orig\_leg} + \\ & \text{orig\_1ch} + \text{orig\_3ch} + \text{orig\_45} + \text{orig\_65} + \text{orig\_cand\_female} \end{aligned}$$

as a linear probability model via `feols()`, clustering standard errors by respondent (`responseid`). Each non-intercept coefficient is an AMCE: the average marginal effect of that attribute, relative to its dropped reference level, on the probability a candidate profile wins its pairwise comparison.

### 3 Method

Point estimates and cluster-robust (“CRV1”) standard errors come directly from `feols()`. Anytime-valid p-values and confidence-sequence radii are then computed from those same estimates using the e-value construction of the parent paper, with the reference degrees of freedom and the process index both set to  $G - 1$  and  $G$  respectively, where  $G$  is the number of clusters (respondents) observed so far – not the row count – since clusters, not rows, are the unit that accumulates sequentially here.

The tuning parameter  $g$ , which controls the shape of the confidence sequence, is fixed once in advance at the value that minimizes the CS radius at the dataset’s actual final size,  $G = 2,105$ . This gives  $g^* = 255.77$ . Critically,  $g^*$  is computed once from the planned final sample size and held fixed throughout sequential monitoring below; re-optimizing  $g$  as data accumulate would invalidate the anytime-valid guarantee, since  $g$  would then depend on how the monitoring unfolds.

All significance levels below use  $\alpha = 0.05$ .

## 4 Results

### 4.1 Confidence sequences over time

Figure 1 shows every attribute’s confidence sequence narrowing steadily as more respondents are observed, while remaining valid to inspect at any point along the way – there is no multiple-looks penalty for having watched, say, the experience effect firm up over the first few hundred respondents before checking in on the candidate-gender effect. The one exception is Age 65 (ref: 29), whose confidence sequence still includes zero at the full sample size ( $-0.049$  to  $0.021$ ); every other attribute’s confidence sequence excludes zero by  $G = 2,105$ .

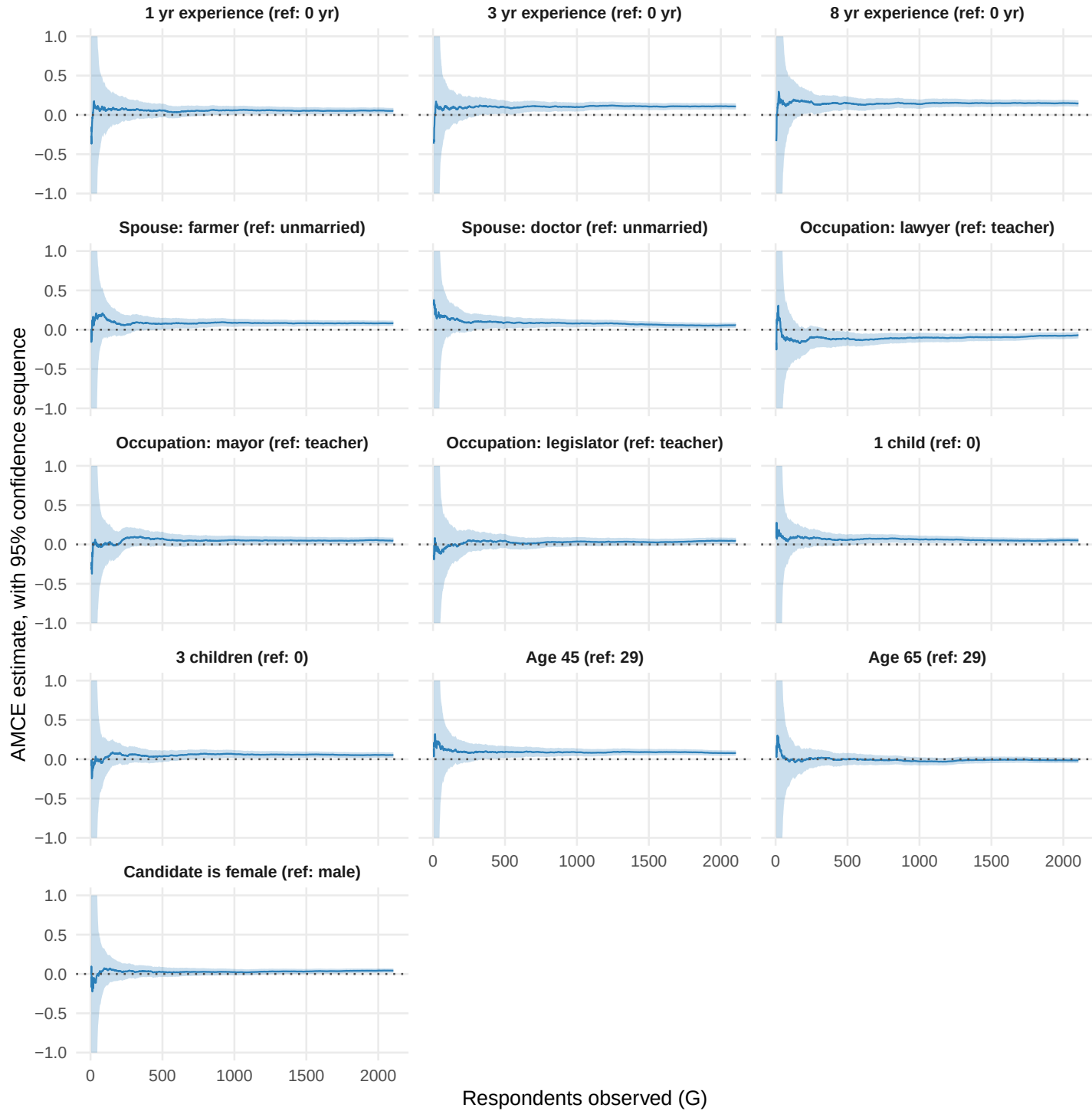


Figure 1: Anytime-valid 95% confidence sequence for each AMCE as respondents accumulate. Shaded regions clipped at  $\pm 1$ ; very early on (roughly  $G < 42$ ) the true radius is unbounded, which is expected: a confidence sequence must sometimes be uninformative to guarantee valid coverage at every sample size, not just at a single pre-planned  $n$ .

## 4.2 e-values over time

Figure 2 shows the corresponding e-value trajectories. 12 of the 13 attributes cross the  $1/\alpha$  rejection threshold by the end of the sample; the exception is Age 65 (ref: 29), whose e-value never separates from neutral evidence (it is 0.66 at the final sample size), consistent with its confidence sequence including zero throughout (Figure 1). Among the 12 attributes that do cross, the number of respondents needed to first do so varies substantially: from  $G = 178$  for 8 yr experience (ref: 0 yr) to  $G = 1,832$  for Occupation: legislator (ref: teacher). Candidate gender – the attribute of central interest here – first crosses at  $G = 1,397$ , i.e. roughly 66% of the way through the sample, making it the 11th of 12 crossing attributes to resolve. The trajectories are not always monotonic once above the threshold: 9 of the 12 crossing attributes, including candidate gender, dip back below it at least once after their first crossing. This is expected of e-processes and is harmless under the relevant decision rule, which is to stop and reject the very first time the threshold is crossed, rather than requiring it to stay crossed forever.

## 4.3 Fixed- $n$ interval vs. confidence sequence, at the final sample size

Figure 3 makes explicit the price of anytime validity at a single, fixed sample size: the two methods share the same point estimate (they are computed from the same model fit), but the confidence sequence is visibly wider than the conventional interval for every AMCE. That extra width is exactly what buys the ability to monitor the study continuously, as in Figures 1 and 2, and still report a valid interval or rejection decision at whatever point the analyst chooses to stop – something the conventional interval does not license. For candidate gender specifically, both intervals exclude zero at the final sample size, and both are centered on the same estimate, 0.042: female candidates are, if anything, slightly more likely to win a pairwise comparison than otherwise-identical male candidates in this sample.

# 5 Discussion

On this real conjoint dataset, 12 of the 13 attributes – years of experience, spousal status, occupation, number of children, age 45 (but not age 65), and candidate gender – have a confidence sequence that excludes zero and an e-value that comfortably clears the rejection threshold by the end of data collection. As in the companion case study, the anytime-valid approach trades away some precision at any single fixed sample size for the ability to monitor these effects continuously and still report a valid interval or rejection decision at whichever point an analyst actually chooses to stop, rather than only at one pre-registered  $n$ .

**Candidate gender.** The coefficient on `orig_cand_female` is the attribute of central substantive interest for this case study, since the motivating question behind the underlying experiment is whether candidate gender affects voter support. Its anytime-valid point estimate is +0.042: holding experience, spousal status, occupation, number of children, and age fixed, a female candidate is estimated to be about 4.2 percentage points *more* likely to win a pairwise comparison against an otherwise-identical male candidate among 2014 general-population voters, and this effect is anytime-valid significant ( $e = 6.8 \times 10^3$  at the final sample size). This is consistent with the original study’s headline finding of no outright hostility toward female candidates among the general public – if anything a modest advantage here – rather than evidence of straightforward voter bias against women. At the same time, the candidate-gender effect is both comparatively small in magnitude (about 29% of the size of the largest AMCE in this study, 8 yr experience (ref: 0 yr), at 0.145) and comparatively slow to resolve under continuous monitoring (the 11th of 12 attributes to cross the

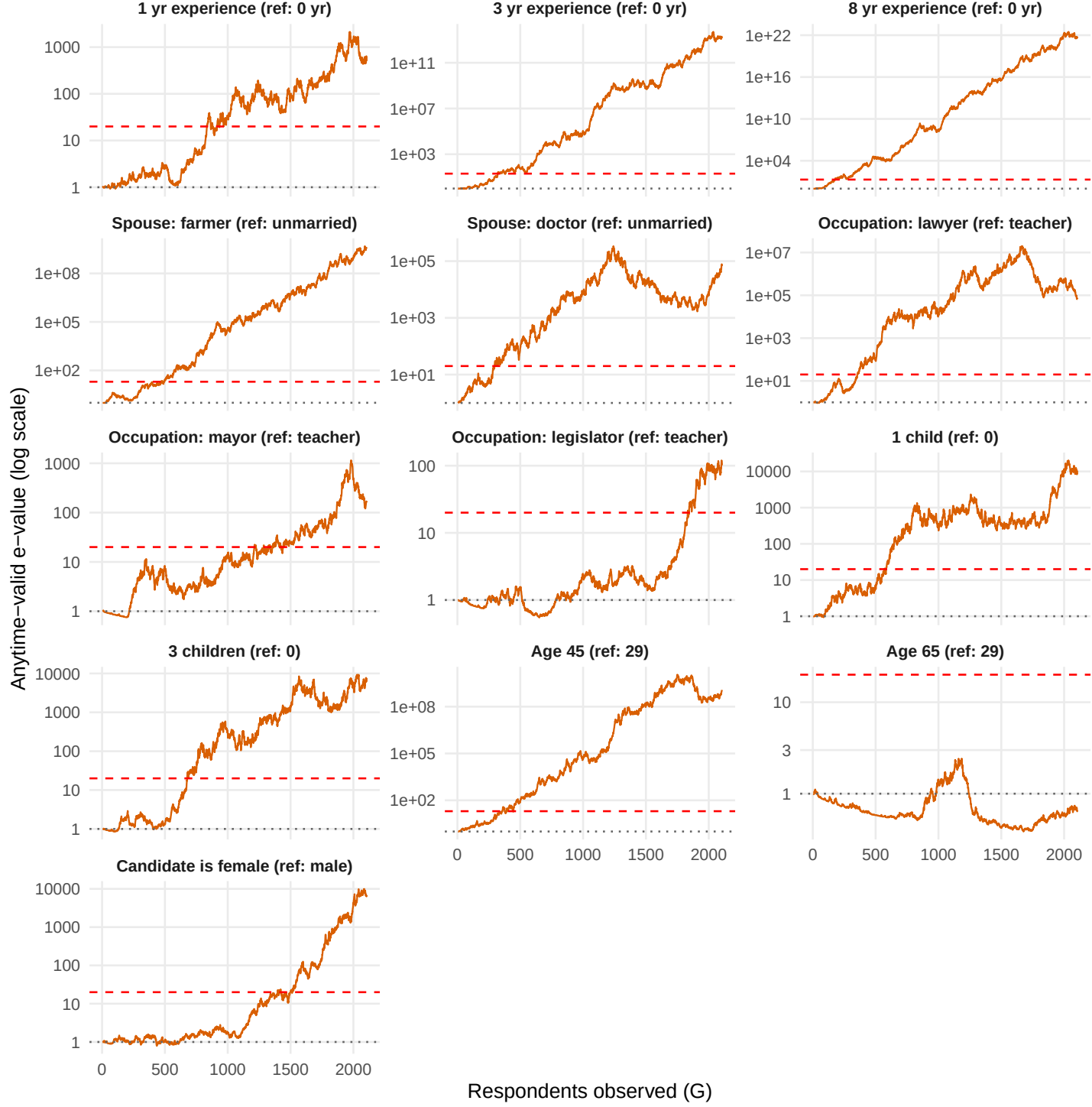


Figure 2: Anytime-valid e-value for each AMCE as respondents accumulate (log scale). The dashed red line is  $1/\alpha$ , the threshold an e-value must cross to reject the null of no effect at level  $\alpha$ ; the dotted grey line is  $e = 1$  (neutral evidence).

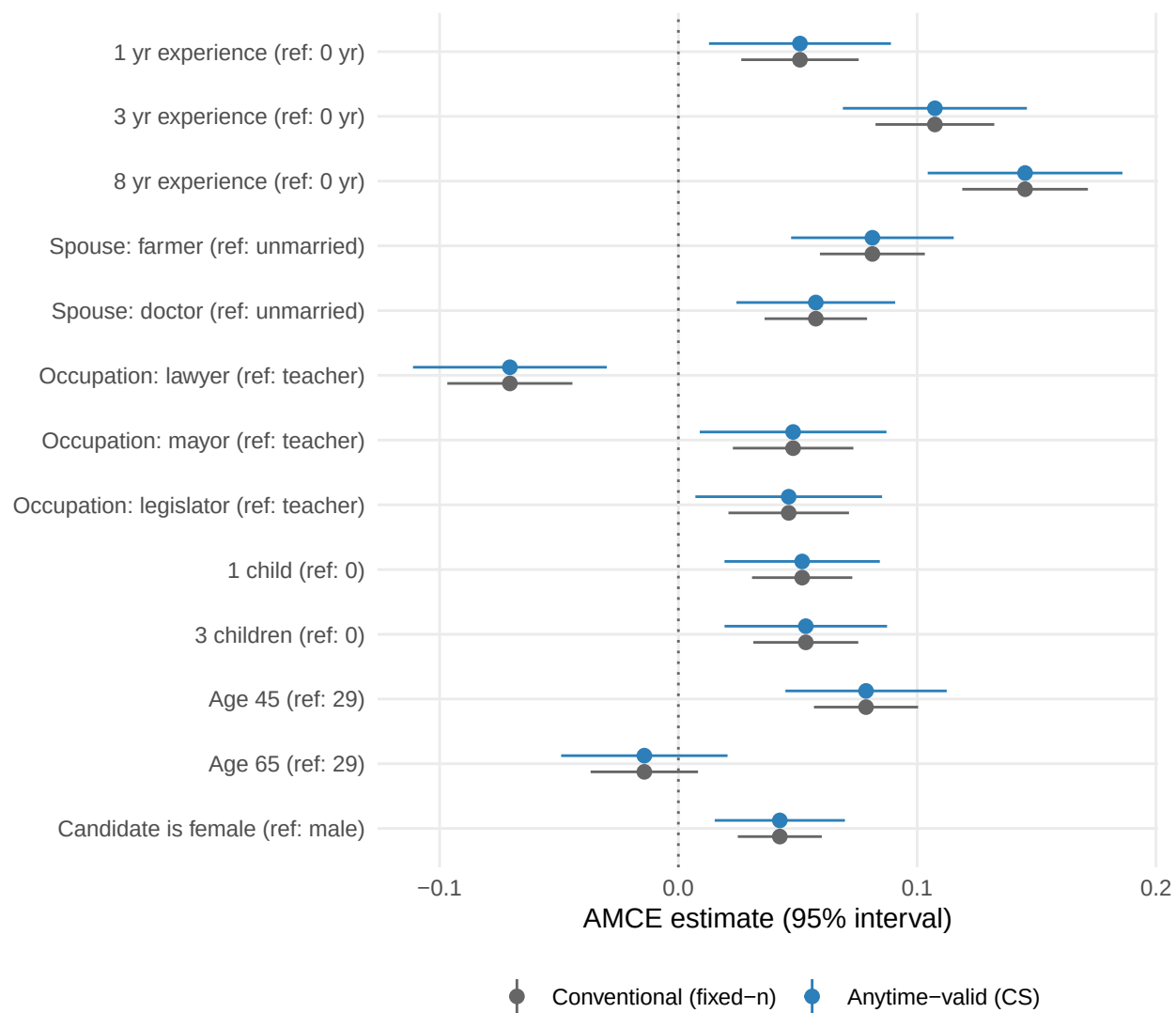


Figure 3: Final AMCE point estimate with a conventional fixed- $n$  95% confidence interval versus the anytime-valid 95% confidence sequence, both computed from the same final model fit ( $G = 2,105$ ).

e-value threshold, at 66% of the way through the sample). A study designed and powered only around the large, fast- resolving experience or occupation effects could easily be underpowered to say anything conclusive about the smaller, slower gender effect at the same fixed sample size – exactly the scenario anytime-valid monitoring is designed for, since it lets data collection continue, with a running valid answer at every point, until whichever effect the analyst cares about has actually resolved.

Two further caveats are worth flagging as natural extensions. First, this case study fits a single additive AMCE model and so cannot speak to the original study’s central “double bind” finding – that traditionally gendered traits such as marital status, children, and occupation are rewarded or penalized *differently* depending on candidate gender. Testing that would require anytime-valid inference on interaction terms (e.g. `orig_cand_female`  $\times$  occupation), which the e-value and confidence-sequence construction used here extends to directly but which we leave to future work. Second, this case study deliberately restricts to the 2014 general-population voter sample under the original attribute set; the 2018 study also fielded a parallel sample of political elites/legislators and a 2017 legislator-only replication with a modified attribute set, both of which would make natural additional anytime-valid case studies, potentially with different conclusions given that political elites and the mass public need not react to candidate gender the same way.

## 6 References

- Teele, D. L., Kalla, J., and Rosenbluth, F. (2018). The Ties That Double Bind: Social Roles and Women’s Underrepresentation in Politics. *American Political Science Review*, 112(3), 525-541.
- Teele, D. L., Kalla, J., and Rosenbluth, F. (2018). Replication Data for: The Ties That Double Bind: Social Roles and Women’s Underrepresentation in Politics. *Harvard Dataverse*. doi:10.7910/DVN/FVCGHC.
- Lindon, M., Ham, D., Tingley, M., and Bojinov, I. (2026). Anytime-valid linear models and regression adjusted causal inference in randomized experiments. *Journal of the American Statistical Association*. arXiv:2210.08589.
- Molitor, D., Gosciak, J., and Lindon, M. (2026). Anytime-Valid Inference in Conjoint Experiments.